

1. Administrative & Study Identification

Full Study Title: A first-in-human, phase I, multicenter, open-label study of MGY825 single agent with a dose escalation and a dose expansion in adult patients with advanced non-small cell lung cancer **Short Title/Acronym:** Study of MGY825 in Patients With Advanced Non-small Cell Lung Cancer **Protocol Number:** CMGY825A12101 **NCT Number:** NCT05275868 **Sponsor Name:** Novartis **Investigational Product:** MGY825 **Phase of Development:** Phase I **Medical Monitor:** Novartis Clinical Operations Lead / Principal Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To determine the safest and most effective dose, characterize the safety and tolerability of MGY825, and establish the Maximum Tolerated Dose (MTD) or Recommended Phase 2 Dose (RP2D). **Secondary Objectives:** To evaluate the pharmacokinetic (PK) profile of MGY825 and assess preliminary anti-tumor activity (e.g., tumor shrinkage, Overall Response Rate).

2.2 Background & Rationale To address the unmet medical need and offer better targeted treatments for patients with advanced (metastatic or unresectable) non-small cell lung cancer (NSCLC) who have already exhausted standard therapeutic options. The study is particularly focused on identifying efficacy in subsets of patients harboring specific genetic alterations, such as NFE2L2, KEAP1, or CUL3 mutations, which may dictate how well the tumor responds to the investigational agent.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation and dose expansion cohorts) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 140 evaluable patients **Number of Sites:** Multicenter **Estimated Study Start:** 2022 **Estimated Primary Completion:** 28-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 to ≤ 100 years. 2. Histologically or cytologically confirmed diagnosis of advanced

(metastatic or unresectable) NSCLC (Group 1: strictly NFE2L2/KEAP1/CUL3 mutant NSCLC; Group 2: irrespective of NFE2L2/KEAP1/CUL3 mutation status). 3. Ability to understand and sign informed consent prior to participation.

4.2 Exclusion Criteria 1. Presence of symptomatic CNS metastases, or CNS metastases requiring local CNS-directed therapy or increasing doses of corticosteroids within 2 weeks prior to study entry. 2. Impaired cardiac function or clinically significant cardiac disease (e.g., congestive heart failure requiring treatment [NYHA Grade ≥ 2], uncontrolled hypertension, clinically significant arrhythmia, or QTcF > 470 msec). 3. Known active COVID-19 infection or inability/unwillingness to swallow capsules as per dosing schedule.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MGY825 administered at escalating dose levels to determine MTD/RP2D. **Route of Administration:** Oral (Capsules). **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Stable palliative/supportive care (e.g., corticosteroids for treated brain metastases must be neurologically stable for 4 weeks post-treatment and at a dose ≤ 10 mg/day prednisone or equivalent for at least 2 weeks before administration). **Prohibited:** Other investigational agents and concurrent systemic anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Mutational Testing (NFE2L2/KEAP1/CUL3), ECG, Labs
Baseline	Day 1	First Dose, Vital Signs, PK Baseline
Treatment (Cycle 1)	Days 1-28	Safety Labs, PK Assessments, DLT Evaluation Window
Treatment (Subsequent)	Day 1 of each Cycle	Safety Labs, Efficacy Assessment (Tumor Imaging), Dispense IP
End of Study	30 days post-last dose	Final Vital Signs, Safety Review, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of dose-limiting toxicities (DLTs) defined as an adverse event or abnormal lab value of CTCAE grade ≥ 3 not primarily related to disease progression occurring during the first 28 days of treatment. Incidence and severity of Adverse Events (AEs) and Serious

Adverse Events (SAEs). **Measurement Method:** Clinical assessment and standard laboratory evaluations graded by CTCAE.

7.2 Secondary & Exploratory Endpoints Pharmacokinetic profile of MGY825 (C_{max}, AUC). Preliminary anti-tumor activity including Overall Response Rate (ORR) and Duration of Response (DoR) based on tumor shrinkage.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by the investigator, grading all AEs via CTCAE. Focus on DLTs within the first 28 days. **Serious Adverse Events (SAEs):** Routine 24-hour reporting protocol for SAEs to the sponsor. **Data Monitoring Committee (DMC):** Internal safety review committee governing dose escalation decisions and cohort progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who received at least one dose of MGY825), Dose-determining Set (for MTD evaluation). **Sample Size Justification:** Target $N = 140$ based on standard Phase I dose escalation and expansion statistical frameworks to adequately assess safety, tolerability, and initial efficacy signals in biomarker-selected cohorts. **Handling of Missing Data:** Standard descriptive statistics with missing data excluded from safety and PK calculations unless otherwise specified by imputation rules for survival endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, accurate AE reporting, and proper IP accountability. **Audit Trail:** Source data verification and eCRF tracking to validate mutational status documentation and DLT observations.

11. Study Identifiers & Governance

Primary ID: CMGY825A12101 **Registry Number:** NCT05275868 **Sponsor/Lead Organization:** Novartis **Study Title:** Study of MGY825 in Patients With Advanced Non-small Cell Lung Cancer **Official Regulatory Title:** A first-in-human, phase I, multicenter, open-label study of MGY825 single agent with a dose escalation and a dose expansion in adult patients with advanced non-small cell lung cancer

12. Clinical Framework (Oncology Specific)

Primary Condition: Advanced (metastatic or unresectable) Non-Small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Histologically or cytologically confirmed; genetic subtyping for NFE2L2, KEAP1, or CUL3 mutations **Medical Methodology:** Targeted Oral Monotherapy **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Phase I Dose Escalation followed by Dose Expansion **Education Protocol (HCP):** Site initiation visits and ongoing investigator training for precise DLT reporting and PK sampling. **Education Protocol (Patient):** Thorough informed consent process focusing on early phase trial risks and rigid dosing schedules. **Quality Audit Mechanism:** Central review of mutational testing data and independent safety data monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression, unacceptable toxicity, or trial completion (Estimated 2027). **Onsite Visit Cadence:** Intensive during Cycle 1 (Days 1-28) for DLT assessment, transitioning to standard oncology cycle visits (e.g., Q3W or Q4W). **Remote Monitoring Frequency:** Supplemental to in-person safety and survival follow-ups. **Checklist Review Interval:** Continuous tracking of AEs and concomitant medications at every clinical encounter.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				